

Total No. of Questions : 4]

SEAT No. :

PE-555

[Total No. of Pages : 2

[6578]-28

S.E. (IT) (Insem)

LOGIC DESIGN & COMPUTER ORGANIZATION

Information Technology

(2019 Pattern) (Semester - III) (214442)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve questions Q.1 or Q.2, Q.3 or Q.4
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data if necessary.

UNIT - I

Q1) a) Explain following Characteristics of digital IC (TTL/CMOS): [5]

- i) Fan in
- ii) Fan Out
- iii) Noise Margin
- iv) Figure of merit

b) Represent $(+60)_{10}$ & $(-60)_{10}$ in - Sign-Magnitude form, 1's complement form and 2's Complement form in 8- bit number format. [5]

c) Minimize the following function using K map in POS Form.

$$f(A, B, C, D) = \pi M(0, 1, 3, 5, 6, 7, 10, 14, 15) \quad [5]$$

OR

Q2) a) Explain with help of suitable circuit diagram operation of 2-i/p CMOS NAND gate? [5]

b) Represent $(-0.8125)_{10}$ in Single precision and Double precision format. [5]

c) Convert the given expression into standard SOP form and simplify using K Map. [5]

$$Y = A + BC + ABC$$

P.T.O.

UNIT - II

- Q3)** a) Design and Implement 03 bit Binary to Gray Code convertor using basic gates. [5]
- b) Draw the 9's Complement circuit using IC 7483. [5]
- What are the rules of BCD Subtraction using 9's complement method.
- c) Design Full Adder (FA) using IC 74153. [5]

OR

- Q4)** a) Design 03 bit Binary to Gray Code convertor circuit using Decoder IC 74138. [5]
- b) Design and Implement 8:1 MUX using 4:1 MUX and Implement given function $F(X, Y, Z) = \sum m(1, 3, 4, 7)$. [5]
- c) State the rules of BCD Addition and Perform BCD Addition of $(75)_{BCD} + (35)_{BCD}$. [5]

